

MODULE 9.2

Antimalware Protection

Cybersecurity Essentials

Topics Covered:

- 9.2.1 Endpoint Threats • IoT • Malware Statistics
- 9.2.2 Endpoint Security • External vs Internal LAN Threats
- 9.2.3 Host-Based Malware Protection • Antivirus / Firewall / Security Suites
- 9.2.4 Network-Based Malware Protection • AMP / ESA / WSA / NAC
- 9.2.5 Check Your Understanding

9.2.1 Endpoint Threats

What is an endpoint — and why are there so many targets now?

An "endpoint" is any host on the network that can access or be accessed by other hosts. This includes computers, servers, and now — thanks to the Internet of Things (IoT) — security cameras, controllers, smart light bulbs, and appliances.

Traditional Endpoints

Laptops, desktops, servers, IP phones
— always-on targets for malware.

IoT Devices

Cameras, controllers, smart
appliances. Often poorly secured and
ignored.

VPN / Remote Devices

Devices on public/home networks that
tunnel into corporate networks — and
can bring malware with them.

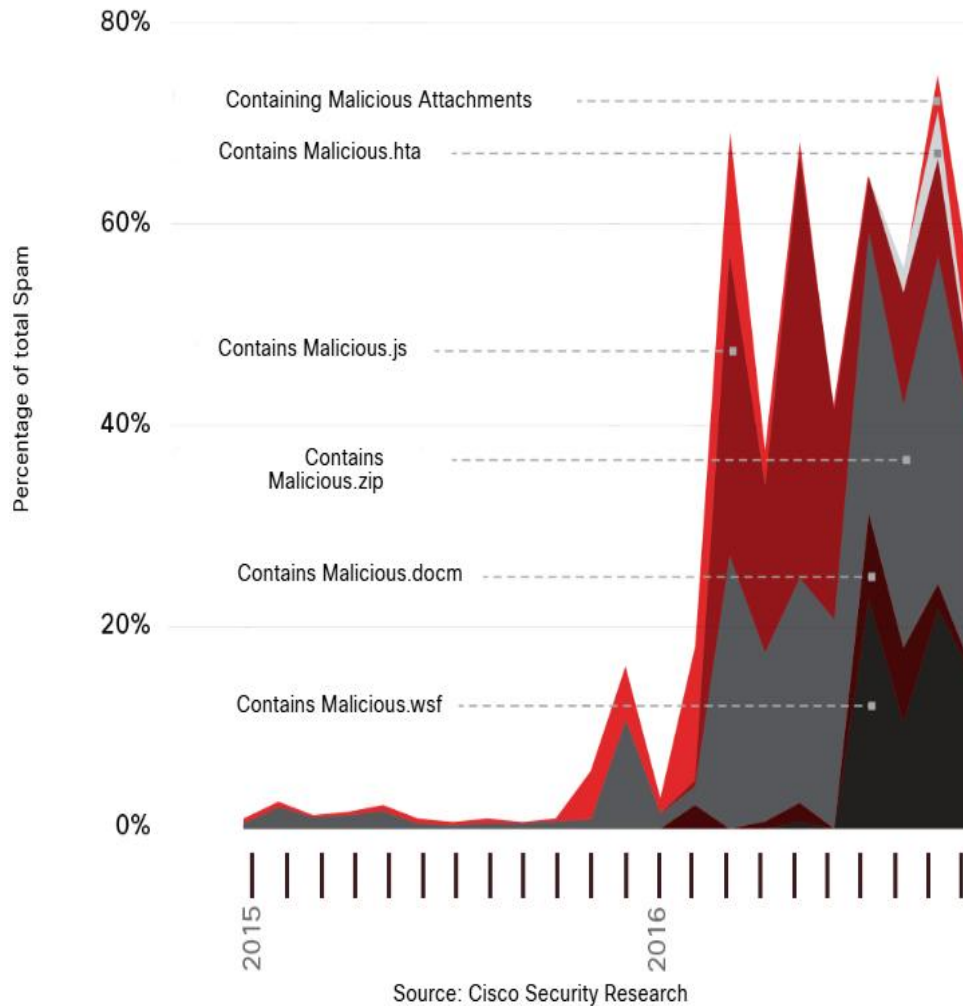
Cloud also expands enterprise network boundaries to internet locations outside an organization's control.

Why Malware Remains a Major Challenge — Key Statistics

Every 11 sec	A new organisation falls victim to ransomware (by 2021 — Cybersecurity Ventures)	\$6 Trillion/yr	Global economic cost of ransomware attacks by 2021	8 Million	Cryptojacking attempts to steal system resources observed in 2018
8–10% Spam	Global spam labelled malicious (2016–2017 spike)	4.8 → 14.2	Avg cyberattacks per macOS device projected to rise 2018–2020	< 24 Hours	Many malware types change features to evade detection within a day

9.2.1 Malicious Spam Percentage

Cisco Security Research: rise of malicious email attachments 2015–2016



Attachment Types Found in Malicious Spam

.hta files	> 70%
.js files	~50%
.zip files	~40%
.docm files	~25%
.wsf files	~15%
All malicious attachments	Rising from 0→70%+

Source: Cisco Security Research — spam volume rose from 0.5K/sec (2012) to over 3K/sec (2016)

9.2.2 Endpoint Security

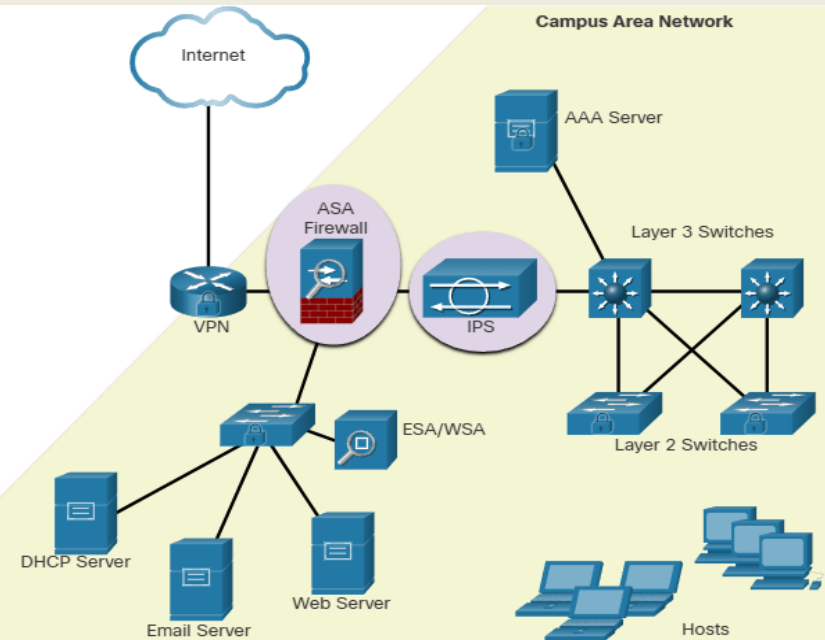
Protecting both the network perimeter and the internal LAN

External Attacks (Perimeter Threats)

- DoS attacks — degrade or halt public access
- Web server breach — deface an org's web presence
- Data server breach — steal confidential information

Perimeter Security Devices

- Hardened VPN router — secure remote access
- ASA Next-Generation Firewall — traffic inspection
- IPS Appliance — intrusion prevention
- AAA Server — authentication, authorization, accounting



Internal LAN Security — Two Elements to Secure

Endpoints

Laptops, desktops, printers, servers, IP phones — all susceptible to malware-related attacks.

Network Infrastructure

Switches, wireless devices, IP telephony — vulnerable to MAC overflow, spoofing, DHCP, LAN storm, STP, and VLAN attacks.

9.2.3 Host-Based Malware Protection — Antivirus & Antimalware

Software installed directly on the device to detect and neutralise threats

As network perimeters expand through VPNs, mobile devices, and cloud, host-based protection becomes critical. These tools work directly on the endpoint, independent of network controls.

Three Antimalware Detection Approaches

Signature-Based

Recognises specific characteristics of known malware files — like a fingerprint database.

*Fast and reliable for known threats.
Cannot catch brand-new malware.*

Heuristics-Based

Looks for general features shared across many types of malware — suspicious behaviour patterns.

Can catch new or slightly modified malware. May produce false positives.

Behavior-Based

Analyses what a program actually DOES at runtime — flags suspicious actions like registry writes or network calls.

Best at catching zero-day threats. More resource intensive.

Agent-Based vs Agentless Antivirus

Agent-Based (Traditional)

Runs on every protected machine. Direct, real-time protection. Can be resource-heavy on virtualised systems.

Agentless (Centralised)

Scans hosts from a central system — popular in virtualised environments (e.g. VMware vShield). Reduces per-VM overhead.

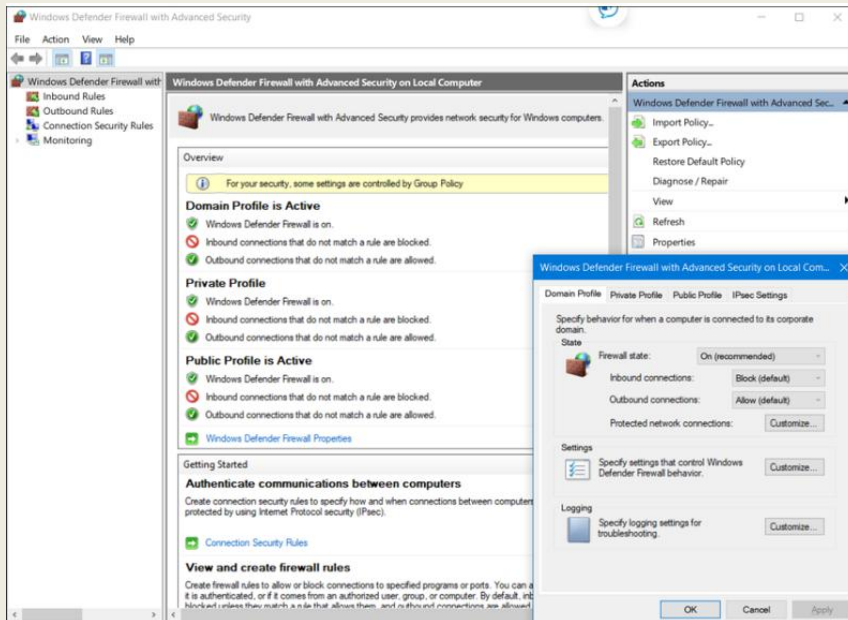
9.2.3 Host-Based Firewall & Security Suites

Layered host-level defence — firewalls and full security suites

Host-Based Firewall

Installed on the device — restricts incoming and outgoing connections to only those initiated by that host. Can also prevent the host from becoming infected or spreading malware.

- Windows Defender Firewall with Advanced Security (built-in)
- Linux iptables — command-line packet filtering
- TCP Wrappers — host-based access control for Linux



Host-Based Security Suites

Recommended for both home and business networks. A full suite combines multiple protection layers in one package — the "defence in depth" approach.

Antivirus

Scans and removes malicious files

Anti-Phishing

Blocks fake sites designed to steal credentials

Safe Browsing

Warns before accessing dangerous websites

Host-Based IPS (HIPS)

Monitors and blocks suspicious behaviour

Firewall

Controls inbound/outbound connections

Telemetry / Logging

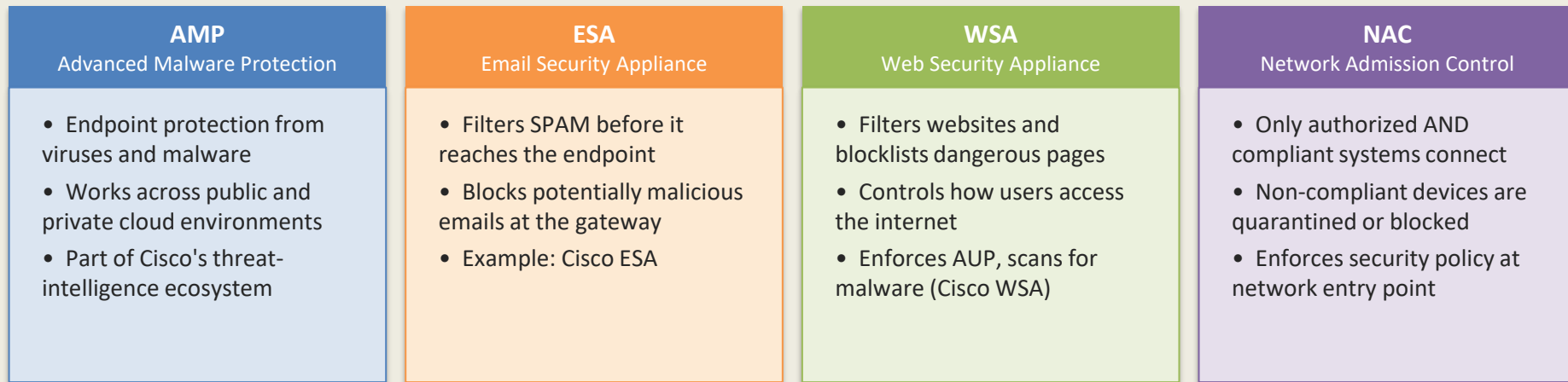
Sends logs to central location for analysis

AV-TEST (av-test.org) provides independent testing and reviews of host-based security products.

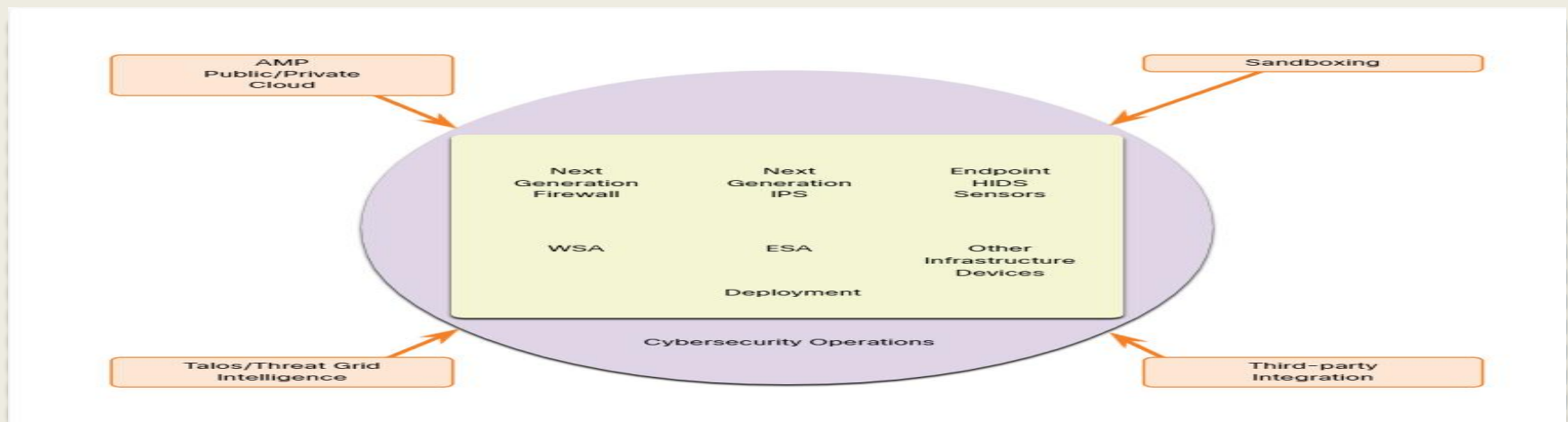
9.2.4 Network-Based Malware Protection

Protecting endpoints at the network level — borderless security architecture

New security architectures address the borderless network by having endpoints use network scanning elements that provide far more scanning layers than a single endpoint could manage alone. These devices also share intelligence with each other to make better decisions.



Cybersecurity Operations Hub — How These Tools Work Together:



9.2.5 Check Your Understanding

Key concept review — answers to the 5 quiz questions

Q1	True or False? Endpoints are hosts on the network that can access or be accessed by other hosts.	TRUE	<i>Endpoints are defined as any host that can access or be accessed on the network — computers, servers, IoT devices, VPN clients, etc.</i>
Q2	Which antimalware approach recognises various characteristics of KNOWN malware files?	Signature-Based	<i>Signature-based detection uses a database of known malware 'fingerprints' to identify threats.</i>
Q3	Which endpoint protection type includes iptables and TCP Wrappers?	Host-Based Firewall	<i>Both iptables (Linux packet filter) and TCP Wrappers are host-based firewall / access control tools.</i>
Q4	What provides filtering of websites and blocklisting to prevent endpoints reaching malicious pages?	Web Security Appliance (WSA)	<i>The WSA (e.g. Cisco WSA) filters URLs, enforces acceptable use policies, and scans for malware.</i>
Q5	Which protection type permits only authorized and compliant devices to connect to the network?	Network Admission Control (NAC)	<i>NAC checks device compliance before granting network access — non-compliant devices are blocked.</i>

Module 9.2 — Summary

Endpoint Threats	Any networked device is an endpoint — IoT, VPNs & cloud ALL expand the attack surface.	Malware Stats	Ransomware attack every 11 sec; \$6T cost/yr; spam malicious attachments hit 70%+.
Endpoint Security	Secure BOTH the perimeter (VPN/ASA/IPS/AAA) AND the internal LAN (endpoints + infra).	Host-Based AV	3 detection methods: Signature, Heuristics, Behavior. Agent-based vs Agentless.
Host-Based Firewall	Windows Defender, Linux iptables, TCP Wrappers — restrict connections per device.	Security Suites	Layered defence: AV + anti-phishing + safe browsing + HIPS + firewall + telemetry.
Network-Based AMP	AMP, ESA, WSA, NAC — protect at the network level, share threat intelligence.	Cybersecurity Ops	NGFW + NGIPS + HIDS + WSA + ESA all feed into a central cybersecurity operations hub.